SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE:Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I K. Sakthivel [192521482] *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: k sakthi

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	2	2	
Concept Mapping Structure	25%	2	2	
Linking & Relationships	20%	2	2	
Application & Analysis	20%	1	1	
Explanation & Clarity	15%	1	1	

Total Marks 16 **/30** ***Signature of the Faculty***

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → → Transport → Network → → Physical

respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✖ → Transport ✖ → Application ✖

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping	25%	Constructs a clear, logical, and well-organized concept map with proper	Concept map is mostly clear with minor structural	Concept map lacks clarity, organization, or	No clear structure or incorrect

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

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CO1 A92
CSA0709
Computer Networks

NETWORK ISSUE ANALYSIS & TROUBLESHOOTING

What the problem means.

Physical Layer :
- The physical connection is fine (no cable / signal)
- Devices can send and receive electrical / optical signals

Data Link layer :
- Frame are being created and delivered
- MAC addressing and local communication

Network layer :
- Packets cannot be routed to other network
- possible cause : wrong IP / gateway.

Transport layer :
- End - to end connection is not established
- possible cause : blocked ports, firewall rules, or transport protocol

Application Layer
- Web services and application cannot communicate
- Cause depends on transport layer, so if also fails

Start from the bottom

Since physical and Data link are working, we can rule out cable, signal and local frame issues

Focus on the Network layer

Check IP address, Subnet mask default gateway and routing table.

Check the transport layer

After fixing the network layer verify if the correct port are open and (TCP/UDP)

Fix the Application layer

Once transport work check the web server, DNS and application settings

Conclusion

The Concept map show that even though the lower layer (physical and Datalink) are working, the failure in the network, Transport and Application layer.

DATALINK → MAC ADDRESS → LOCAL DELIVERY NETWORK → IP ADDRESS → GLOBAL DELIVERY

Data Link

Works at the Data link layer the data is first packed into a frame at the data link layer. This frame helps the network understand where to send the data next on local link

eg: Wifi

MAC ADDRESS

MAC : AA : BB : CC : DD

Identifies the exact device on the local network

The frame uses a MAC address to identify the exact device

Local Delivery Network

Find the next network

The data is delivered within local network using MAC address

The router then checks the IP address

IP Address

IP : 192 : 168 : 1 : 10

Finds the path across networks

The IP address works at the network layer

It identifies the device across different networks and helps router

TOGETHER, THEY ENSURE COMPLETE DELIVERY

Data link — Packs the data

MAC Address — Identifies the devices on the local network

Local Delivery network — Sends to the next network

IP Address — Find the best path across network

Global Delivery — Reaches the final destination